

NIKHIL REDDY LEVAKU

+91-6303252008 | levakunikhilreddy8@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Backend Engineer with production experience building **100+ REST APIs** using **Java**, **Spring Boot**, and **Spring Security** across microservices-based applications. Experienced in **JWT/RBAC** authentication, OTP verification, **Razorpay** payment integration, and asynchronous processing with **Kafka** and **RabbitMQ**. Strong in **PostgreSQL**, **Redis**, **concurrency-safe** transactional workflows, REST API design, and automated testing with JUnit 5 and Mockito.

TECHNICAL SKILLS

Languages & Frameworks	Java, SQL, Spring Boot , Spring Security , Hibernate/JPA
Databases	PostgreSQL, MySQL, Redis, Flyway
Messaging & Architecture	Apache Kafka , RabbitMQ , Microservices, REST API Design
Testing & Tools	JUnit 5, Mockito, Postman, Swagger/OpenAPI, Git, GitHub, Gradle, Maven
Concepts	JWT, RBAC, Transaction Management, Concurrency, Event-Driven Design, Agile/Scrum
Cloud & DevOps	Docker, Docker Compose, Nginx, GitHub Actions (CI), Infisical, Linux, AWS (S3; EC2/IAM basics)

PROFESSIONAL EXPERIENCE

DeepLure Research Pvt Ltd — Backend Engineer May 2026 – Aug 2026

Java · Spring Boot · Spring Security · Microservices · PostgreSQL · Hibernate/JPA · Redis · RabbitMQ · Infisical

- Designed and shipped **100+ production REST APIs** in Spring Boot on a **microservices architecture** across a B2B marketplace and an on-demand services platform, covering authentication, orders, payments, and partner/admin modules.
- Secured APIs with **Spring Security**, **JWT**, and **role-based access control (RBAC)**, with OTP verification via SMS/email.
- Integrated the **Razorpay** payment gateway for orders, wallets, and refunds, using idempotency keys and **optimistic/pessimistic locking** to prevent double-charges and race conditions under concurrent requests.
- Used **RabbitMQ** for asynchronous inter-service notification workflows; managed application secrets with **Infisical**.
- Built real-time chat and support ticketing over **WebSocket/STOMP**, with **Firebase Cloud Messaging** push notifications and third-party SMS integration.
- Engineered **Redis-backed asynchronous workers** for notifications, payment reconciliation, and booking expiry, and reduced database load **~40%** through Redis caching and query optimization.
- Modeled normalized **PostgreSQL** schemas with Hibernate/JPA and versioned **Flyway** migrations; wrote unit and integration tests with **JUnit 5** and **Mockito**; contributed in Agile sprints via code reviews, debugging, and defect resolution.

SmartBridge Educational Services (APSCHE) — Data Analytics Intern May 2025 – Jul 2025

- Cleaned and transformed toy-manufacturing sales datasets using SQL joins, filters, and aggregate queries, and built 3 Tableau dashboards for state-level production and inventory review.

PROJECTS

Flight Booking System Aug 2025 – Jan 2026

Java · Spring Boot · Spring Security · Microservices · PostgreSQL · Redis · Apache Kafka · Docker · Nginx · GitHub Actions | [GitHub](#)

- Refactored a monolithic Spring Boot backend into **4 independently deployable microservices (Auth, Flight, Booking, Notification)**, each with its own Spring Boot application, **PostgreSQL database**, and **Flyway** migrations.
- Implemented **REST-based inter-service communication**: the Booking Service calls the Flight Service to reserve seats, **propagating the user's JWT** so each service independently authenticates and enforces **RBAC** for user and admin roles.
- Made seat reservation **concurrency-safe** with row-level **pessimistic write locks (PESSIMISTIC_WRITE)** inside **@Transactional** methods, preventing overbooking when concurrent requests target the same seats; added seat rollback logic.
- Built an **event-driven** flow with **Apache Kafka**: Booking Service publishes a booking-created event, consumed asynchronously by a dedicated **Notification Service**, decoupling notifications from the booking transaction.
- Added **Redis caching** in the Flight Service; validated requests with **Jakarta Bean Validation**, handled errors through centralized exception handling, and covered services with **JUnit 5/Mockito** tests.
- Containerized each service with multi-stage **Docker** builds and ran the stack with **Docker Compose** behind an **Nginx reverse proxy** (path-based routes to Auth, Flight, Booking); added a **GitHub Actions** workflow for Gradle build/test.

User Management System Oct 2024 – Dec 2024

Java · Spring Boot · Hibernate/JPA · MySQL · REST API | [GitHub](#)

- Built a CRUD REST API with dynamic search and filtering, request validation, and centralized exception handling.
- Modeled a normalized 6-table MySQL schema with JPA entity mappings; tested and documented all endpoints in Postman.

EDUCATION

Sri Venkateswara College of Engineering , Tirupati	2022 – May 2026
B.Tech, Electronics & Communication Engineering	CGPA: 8.5 / 10

CERTIFICATIONS

Oracle Cloud Infrastructure Foundations Associate — Oracle (Sep 2025) · Problem Solving, Intermediate — HackerRank (Apr 2026) · SQL, Intermediate — HackerRank (Mar 2026) · Java Programming — GeeksforGeeks (Mar 2026)